

Sweet Tarmac
The complete guide

Every feature, numbered, and how it all works: how a road is measured, what your phone does with it, what is shared with other drivers, and what the server does with it.

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Features are numbered 1 to 65. Part 2 is the list; parts 3 to 6 explain how each thing is done.

1. Why this app exists

Some people drive to get somewhere. Others drive because they love to drive: to go through the gears, to flow through corners, to enjoy a machine they admire on a road that lets it breathe. For them the journey is the point, and the destination is simply the place they come back to.

What spoils that drive is never the car. It is the bump nobody warned about, the pothole in the best bend, the manhole cover in the middle of the racing line, broken asphalt, bad cobbles, and a string of traffic lights that kills the flow. Maps know where roads go; they do not know how they feel.

Sweet Tarmac measures how the road feels, with the phone that is already in the car, and turns it into three things: a **map of the roads you have driven, coloured from smooth to rough**; a **count of the bumps, holes and manhole covers** on them; and, above all, the **Loop**: a round trip, of the time or distance you choose, that starts and ends exactly where you are and runs over the smoothest roads known, steering round the things that spoil a drive. What you record can also, if you choose, help other drivers who love the same thing to find the roads worth driving.

This guide lists every feature with a number, and then explains how everything works: how a reading is measured, what stays on your phone and what it is used for, what is shared with others in online mode, and how the server turns many drivers' readings into one smoothness and obstacle map and answers requests with it.

2. The features, numbered

Each feature has a number; the illustrated version of this list is on the website (features page).

Recording a drive

Mount the phone firmly, press one button and drive. Everything below happens while you do.

1. **One-tap recording that keeps going.** A single **Start recording** button starts a foreground service that reads the phone's accelerometer, gyroscope and GPS. It keeps running with the screen off or another app in front; the notification has **Pause** and **Stop**.
2. **Works in any phone orientation.** The app removes gravity and measures only the up-and-down movement of the car, wherever and however the phone is mounted. A firm mount, the same every time, gives the best readings.
3. **A smoothness score for every second of road.** For each GPS second the app takes the **median of 0.2-second pieces** of the vertical movement, **plus the left-right rocking of the car** (an uneven pavement at low speed rocks the car without jolting it), so one pothole does not paint a whole road red, and scales it to what it would be at 50 km/h, so the same road scores the same at 30 or 90 km/h. Extra measures describe the texture: wavy roads, fast vibration (cobbles) and lumpy roads.
4. **Only measured when it can be measured.** Smoothness is recorded **from 5 km/h** (nothing below that; at the lowest speeds the readings are a little less exact). Fixes with a GPS accuracy worse than 20 m are rejected, and so are seconds with too few sensor readings.
5. **Live smoothness panel.** While you drive, a panel at the top shows the level of the road right now (for example *Good 3.4*), your speed and a 60-second chart, refreshed four times a second, in the colour of the level.
6. **Live trace and a map that follows you.** The drive is drawn as you go, in colour, on its own layer. Your older drives fade back so the current one stands out (more strongly on the light map). The map can turn heading-up, and zooms in closer in town and further out at speed.
7. **Hold to stop, pause and resume.** Stopping needs a one-second hold on a filling ring, so a stray tap never ends a drive. Pause freezes the clock, and paused time is left out of duration and speed.
8. **Looks after forgotten trips.** After **30 minutes without moving** the trip ends where the car last moved. If you drive off again while paused, recording resumes by itself after a 10-second countdown unless you tap *Stay paused*.
9. **Pauses when the phone is moved.** If the gyroscope shows the phone being picked up, swung or knocked out of its mount (more than about 50 degrees of net turn within 1.5 seconds, fast), recording pauses and a pop-up explains that the phone must be in a fixed support to measure accurately.
10. **Ignores the phone's own buzzing.** The recording notification never vibrates. A detector recognises the phone's vibration motor (one steady frequency between about 70 and 250 Hz) shaking the sensors and leaves that second out, with any bump it seemed to hit.
11. **Trip summary after every drive.** A card shows distance, time, average speed and the share of each smoothness level, with *Show on map* and *Discard*. It closes by itself after ten seconds unless touched, and the last trip can always be discarded.

12 **Optional spoken warnings.** Say “Rough road ahead” before a stretch you recorded as rough, and “Speed bump ahead” or “Pothole ahead, on the right” before the obstacles the app knows.

The map of your roads

Everything you drive becomes a map of how the road feels.

13 **Five levels on a 0 to 5 scale. Smooth** (4-5), **Good** (3-4), **Fair** (2-3), **Rough** (1-2) and **Very rough** (0-1), the same on the map, the live panel, the charts and the route cards. A colour-blind palette is available.

14 **Each road, in each direction.** Roads are measured in cells of about 17 m, separately for each direction of travel (headings within 45 degrees). On a stretch the **latest drive counts fully**, each earlier one fades with age (a month about 37%, three months 5%), and only the last 10 drives count, so a freshly resurfaced road turns green quickly.

15 **Light, dark and auto map.** The layout is light or dark, the map is light, dark or switches at sunset (computed on the phone from where you are). Map tiles come from OpenStreetMap, MapTiler or Stadia, as you choose.

16 **Tap a road to see what is known.** A pop-up shows the level in each direction, how many passes and measurements, when it was last driven, which cars drove it, the surface type (wavy, cobbles, lumpy) and what the other direction is like.

17 **Your numbers at a glance.** When idle, three figures in a gold frame show the **kilometres of road** you have recorded (each direction counted separately), the **obstacles per km** and the **change against the previous 30 days**. Tap them for details and a six-month chart.

18 **Trips list.** Every trip can be shown on the map (all others turn grey), moved to another car, or deleted with Undo.

Bumps, holes and manhole covers

The things that spoil a drive, found, counted and kept off your route.

19 **Finds speed bumps, holes and manhole covers.** Short knocks are stored as events and judged against a bar that depends on speed and on your car's stiffness. The app pairs the knock of the front and rear axle, works out length and height, and uses the gyroscope to tell which side was hit and whether the car rolled or pitched.

20 **Unavoidable or avoidable.** Across repeated drives, an obstacle that is **always hit** is a speed bump or a long table (counted as 2); one that is **sometimes hit**, or that drivers steer round, is a hole or cover you can avoid. Once seen only once it stays “unconfirmed” until you drive there again.

21 **Notices when you steer round something.** On a straight road, a quick move 0.3 to 2.2 m sideways and back, rebuilt from the gyroscope, is read as avoiding a hole or cover.

22 **A quick question for big hits.** After a very deep hit a card with one **Confirm** button appears for three seconds. No tap means “no”; it is decided on the phone, so it works with the screen off.

23 **Ignores car parks and ramps.** Knocks in the first and last 90 seconds at low speed, or in sharp turns, are filtered out as garage ramps and kerbs.

24 **Mark them yourself, in gold.** Two round gold buttons on the Record tab mark a **speed bump** or a **hole / manhole cover where the car is now** (the last GPS fix moved on by speed and heading). Each has its own sound, and your marks are drawn in gold above anything else.

25 **Show or hide them, and choose the sources.** Two switches (both off at first) show or hide detected bumps and holes on the map; off removes them from the colour key, from the Avoid menu and from the hole question too. Your own gold marks always show. You can also switch off speed bumps from other sources (OpenStreetMap, the server) and holes from the server, to use only yours.

26 **Speed bumps and traffic lights from OpenStreetMap.** Known bumps are downloaded in map tiles and cached for a month; traffic lights show from zoom level 15. Your own labels always win over them.

Cars and calibration

A stiff classic and a soft saloon should call the same road smooth.

27 **Up to twelve cars.** Each car has a name (up to 12 characters), a colour from **twelve clearly different ones**, and its own suspension setting. Every reading remembers which car made it; you can switch car with one tap, even while recording.

28 **A suspension scale you can feel.** Stiffness runs from 1.0× to 6.5×, with **Very soft, Soft, Normal (2.5×), Sporty, Stiff, Kart and Rock hard**, and a pair of − / + buttons on the Record tab (with a short falling or rising sound) to fine-tune on the road. Changing it re-colours all your old drives instantly.

29 **Guided calibration. Calibrate my car** asks what the car is like, then fits the setting from roads you know are smooth or rough (pick on the map, use the last 30 seconds, or long-press a road). With a smooth and a rough reference the spread between levels is fitted too.

30 **Warns when one of your cars is off.** With more than one car, if one constantly reads differently from the others on the same roads, the app tells you which one is probably not calibrated, and offers to calibrate it now. Its readings are adding bad data until then.

Smooth routes, A to B

The same roads, chosen for how they feel.

31 **Alternatives compared on your data.** The router (OSRM, openrouteservice or GraphHopper) gives alternatives and the app adds forced detours; every candidate is checked every 15 m against your roads, by direction, with penalties for bumps and holes. Cards show time, distance, share of smooth road, bumps and holes.

32 **Fastest or smoothest.** A *Smooth* slider sets how much feel matters against time. Roads you have not driven count as average.

33 **Search, pins and the map menu.** Search with suggestions for From and To; long-press the map for *Route here*, *Start from here*, a reference road for your car, or to save the place. The pop-up opens in the middle of the visible map so it never lands behind a button.

34 **Home and favourites.** A Home button and up to 12 favourites sit on the Route tab for one-tap planning; **long-press a favourite to rename it**.

35 **The chosen route stands out.** When results are shown, the recorded roads fade back and the chosen route is drawn thicker with a white halo.

36 **See progress, and cancel.** While the phone or the server works, a box shows what is happening, the seconds gone and a progress bar, with a **Cancel** button that drops the request.

The Loop

"Take me for a drive, and bring me back." The reason the app exists.

37 **A round trip by time or distance.** Choose a time (10 minutes to 2 hours) or a distance (up to 200 km / 125 miles) and a tolerance of ±5 to 25%. The Loop starts from where you are, with no destination to enter.

38 **Starts and ends exactly where you are.** Whatever the router returns, both ends of every loop are put back on your position (a route that starts or ends more than 1 km away is thrown out). The map draws the loop from the Start / finish flag.

39 **Over roads that are known and smooth.** The recorded cells form a graph; the planner searches it for a loop of the right size, hands the router a few via points, and requires at least **95% known road** whenever that is possible.

40 **Always a loop with the least unknown road.** Even with little recorded road, the planner also tries **there-and-back** loops, out along recorded roads and back along the same ones, so that the loop with the least unknown road is always offered.

41 **A strict order of priorities.** 1) 95 to 100% known road, the more the better; 2) the smoothness you asked for; 3) the time or distance within your tolerance; 4) as few of the things you avoid as possible. Loops outside the tolerance only fill up to three results.

42 **What to avoid.** Choose any of **traffic lights, speed bumps, big holes and manhole covers, zebra crossings and tolls**. Each light, crossing, toll, bump or hole is worth a 250 m detour in the search, and each loop lists what it still has.

43 **Explorer: roads you have not driven.** A green switch draws a loop over roads you have **not recorded yet**, for the time or distance chosen, and turns the interface green so you know the mode. Drive it with recording on and the map grows.

44 **Explorer on routes.** The Route tab has an **Explorer** button (at the right end of the favourites line) and, in Options, a share from **10% to 100%**: the part of the ride that must be on roads you have **never recorded**, with any car. The button turns green and says "Find smoothest route and explore 40% of unknown". The routes shown are the fastest ones with at least that much new road; if the road network cannot give that much, it says so and shows the closest.

45 **Loop cards that tell you what you get.** Each card shows time, distance, the share of smooth and good road, the share unknown (when there is some), lights, zebra crossings, bumps, holes and the very rough kilometres. The top loop is tagged *Recommended*; the list folds away to leave the map.

46 **Never fewer than three.** When at least three loops exist, you get at least three, with the unknown share shown on any that is over the 5% limit.

Handing the route to a navigator

The app plans; your favourite navigator guides.

47 **Google Maps that follows the planned route.** Google takes at most nine stops and picks the roads between them itself, so the app adds stops where a check router drifts from the planned route until they agree (at least 97% overlap), and tells you how well it matches.

48 **Choose your navigator: OsmAnd, Organic Maps, Google Maps or any app.** One Navigate button remembers your choice: **OsmAnd** (recommended) or **Organic Maps** open the **exact route** as a GPX track; **Google Maps** gets up to 9 stops that keep it on your roads; other apps take a destination only.

49 **Offers to record the drive.** When you send a route to a navigator, a card asks *Start recording this drive?* with **Yes** and **No**; with no answer in two seconds it is Yes.

Your data

It is yours, on your phone, and it can all be taken with you.

50 **Everything is stored on the phone.** Readings, trips, obstacles, cars, settings and places live in a database on the phone. Offline mode, the default, sends nothing anywhere except requests for maps, routes and place search to the services you choose.

51 **A full backup in one file.** *Back up everything* writes one file with every recording and trip, the obstacles and your corrections, the cars with their calibration, all settings, your user id, your server id and your nickname. Restoring it on a new phone or after reinstalling puts everything back as it was, including your place in the explorers ranking. Keep the file private. (Export GeoJSON is still there for maps and other tools.)

52 **Delete what you want.** Delete the last hours, days or trips, or everything. Your own routing and map keys and the anonymous online id are never put in an export.

53 **No account, just a nickname.** The app makes its own random user id and asks for a nickname when you first open it. No e-mail address and no Google account. The id labels your own data; the nickname is only sent if you switch on online mode.

Online mode: drivers helping drivers

Optional, off at first, anonymous, and deletable.

54 **Anonymous, trimmed uploads.** After each trip the readings are sent with the **first and last 300 m cut**, positions rounded to about 1 m, only the day (not the time), the score already adjusted for your car's calibration, and a random id made on the phone (not your e-mail). Trips under about 800 m send nothing.

55 **Anonymous facts about the trip.** With each trip: the date, Android version, country, language, suspension setting, the number of vehicles you use (never their names), your options from a fixed list (no addresses, places, e-mail or keys) and counts of how often you used each feature. The dialog that switches it on says so.

56 **The server plans on everybody's data.** In online mode, routes and loops are planned by the server with the same algorithm and everyone's data; if it cannot be reached the phone plans on its own and says so.

57 **Bad data is kept out.** On the server, readings far from the middle of the normal curve for their stretch of road are put in quarantine and not used for the averages everybody gets. Nothing is deleted.

58 **Explorers ranking.** Choose a unique nickname and open **Explorers ranking** in the menu: the drivers who registered the most **new road** (road nobody had sent before) come first, with medals for the top three, their nickname and the kilometres. **Highways and freeways do not count** (they are almost all smooth; the aim is to record streets and roads). The **golden steering wheel**, animated and sparkly, goes to the one with the most new **smooth and good road** in the last 30 days, as long as the car is calibrated; if it is not, the app says so and offers the calibration routine.

59 **In your hands.** *Send now* and *Delete my data on the server* sit side by side in Options. Deleting removes everything your device sent, including the trip facts.

Made to be used in a car

Big targets, clear colours, your language.

60 **Six languages, km or miles.** English, Português, Español, Français, Deutsch and Italiano, chosen from the phone's language; distances and speeds in km or miles (and mph).

61 **Accessible by design.** 48 px touch targets, text that follows the phone's font size, a colour-blind palette, light and dark layouts, portrait only, and edge-to-edge screens with the new back gesture.

62 **Sounds you can choose.** Marking a bump and a hole each has its own sound, and the suspension − / + a very short falling and rising one. Options has four choices for each (or off), played when you pick.

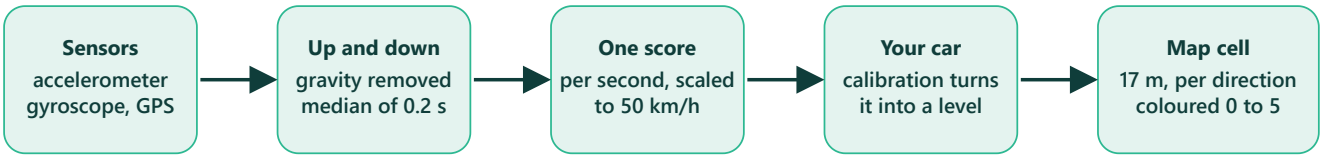
63 **Options that stay out of the way.** Options always opens folded, with the map blurred behind it, and ten sections by what you want to do (My cars, While driving, Map & roads, Routes, Community, App & language, My data, Advanced, About), a search box and a status line on My cars and Community.

Safe by design

64 **Set it up, then drive.** Mount the phone, set the app up before you set off, and never handle it while driving. Big buttons and short holds are there so that you do not have to look.

65 **No ads, no analytics, no tracking.** The app has no advertising and no analytics. The only thing that ever leaves the phone is what the Privacy section of this guide lists.

3. How the app measures a road



3.1 Sensors

While recording, the app reads the accelerometer at the fastest rate the phone allows (often 200 to 400 readings a second), the gravity sensor, the gyroscope and the GPS once a second. A foreground service does this, so it continues with the screen off.

3.2 The up-and-down movement of the car

The phone can be in any position. The app estimates the direction of gravity (a slow average of the accelerometer with a time constant of about 0.8 s, or the phone's own gravity sensor), removes it, and keeps only the part of the remaining acceleration along the gravity axis: how the car moves up and down. A slow average of that signal (about a second) is subtracted, so only the movement caused by the road is left. The same signal feeds the live panel.

3.3 One number for each second of road

Between two GPS fixes the app collects the vertical movement and measures it in pieces of 0.2 seconds. The value used for the road is the **median** of those pieces (a "robust" value), so that a single pothole, which is one violent piece, does not paint a whole second of road red: single hits are handled separately as obstacles. The value is then scaled by a speed model to what it would be at **50 km/h**, so the same road gives the same score at 30, 60 or 90 km/h. This raw score is what is stored.

Rocking left-right. Since v0.9.11 the car's roll (the gyroscope rate round the forward axis, in the band from about 0.3 to 6 Hz, so slow leaning in corners and lane changes is left out) is measured in the same way, as a median over the 0.2-second pieces, and turned into an equivalent vertical acceleration (9 m/s^2 per rad/s above a small noise floor of 0.015 rad/s). It is added to the vertical value as the square root of the sum of squares, so an uneven pavement that rocks the car at 15 to 40 km/h counts as rough even when it does not jolt it up and down. Until the app knows how the phone is mounted, the tilting part of the gyroscope (roll and pitch together, without the turning part) is used instead, counted at 0.7.

Three more numbers describe the texture of the road: the *heave* (slow waves between 0.5 and 3 Hz), fast vibration above about 12 Hz (the signature of cobbles) and the number of standout pieces (a lumpy road). They are used for the surface type shown when you tap a road.

3.4 When a reading is accepted

- **Speed:** only from **5 km/h** (it was 15 km/h from v0.9.7 to v0.9.17). Slower than that nothing is measured; at the lowest speeds the readings are less exact and tend to be a little smoother than the road is.
- **GPS:** fixes with an accuracy worse than 20 m are rejected, and the chain of positions is broken so that a bad fix never becomes the anchor of the next sample; a gap of several seconds in the fixes ends the chain too.

- **Enough sensor readings** must have arrived during the second.
- **The phone must be still in its mount:** if the gyroscope shows the phone being picked up or knocked (net turn above about 50 degrees within 1.5 s with a fast turn rate, not in the first 4 seconds after a start or resume), recording pauses and a pop-up says so. Turns of the car itself (even a tight U-turn is about 1 rad/s), bumps and the shaking of cobbles do not trigger it.
- **The phone's own vibration** (a notification, a call) is recognised as one steady frequency between about 70 and 250 Hz that the road does not produce, and that second is left out. A phone buzz lasts a second or two; a car's own vibration (an old engine, a resonance at some speed) is a steady tone too but lasts: from v0.9.15 only the first two seconds in a row are left out, and when it goes on for three seconds it is treated as the car and nothing is left out for the next minute. The trip summary lists the seconds that were not measured and why (weak GPS, GPS gap, phone vibration, too few sensor readings). The recording notification itself never vibrates. This works on phones with a sensor rate of about 300 Hz or more.

3.5 Where the reading goes

The score is placed at the midpoint of the last two GPS fixes, with the heading of the movement between them, and stored with the time, speed, car and trip. Nothing is smoothed or snapped to roads: the line on the map is what was driven.

3.6 Your car's calibration

A stiff classic and a soft saloon feel the same road very differently, so the raw score is converted with the car's own calibration:

$$\text{calibrated} = P \cdot (\text{raw} / (k \cdot P))^{b} \quad \text{with } P = 0.69$$

k is the stiffness (how many times higher the raw numbers are than for a typical car) and **b** the spread (how quickly roads go from smooth to rough, 0.5 to 2.5). Only the raw score is stored; the conversion is done whenever a road is shown or used, so changing a car's setting re-colours every past drive at once, each drive with the setting of the car that made it.

The stiffness shown to you runs from 1.0× to 6.5× with a normal car at 2.5× in the middle (the number shown is $2.5 \cdot (k / 2.5)^{1.226}$; the presets are Very soft 1.1, Soft 1.7, Normal 2.5, Sporty 3.6, Stiff 4.5, Kart 5.2, Rock hard 6.0). **Guided calibration** asks what the car is like, then fits k from a road you mark as smooth (it should land in the middle of the smooth band) or rough, and, when you have marked both kinds, also fits b so that the two references land on the middle of their bands.

3.7 From score to level

The calibrated score becomes a level from 0.0 to 5.0 (5 is smoothest) by straight lines between these points:

calibrated score	0	0.4	0.7	1.0	1.8	3.0
level	5	4	3	2	1	0

and the levels form five bands, each one wide: **Smooth** 4–5, **Good** 3–4, **Fair** 2–3, **Rough** 1–2 and **Very rough** 0–1. The same function is used on the phone's screen and in its engine.

4. What the app does with your data, on your phone

4.1 The roughness grid

All readings are kept in a database on the phone and in a grid of cells of about 17 m (0.00015° of latitude by 0.0002° of longitude). In each cell the app keeps the *passes*: one pass is one drive (trip) of one car through the cell in one direction (headings within 45 degrees count as the same direction), with the average raw score of that pass and its day.

4.2 One value per road, per direction

To get the value of a cell for a direction, each pass is converted with its own car's calibration and the passes are blended: the **latest drive counts fully** and each earlier one is weighted by $e^{-\text{days older} / \tau}$ with $\tau = 90 / \ln 20 \approx 30$ days (the same day 100%, a month about 37%, three months 5%), and **only the last 10 drives** count. This single rule is used everywhere: the map, route and Loop scoring, spoken warnings, the road information on tap and the statistics. A road that was resurfaced turns green after one new drive.

4.3 “Known” roads

A cell is *known* when it has been driven in any direction (most recordings are one way round, and the other way is the same road). Known roads are what the Loop is built from; unknown roads are assumed to be of average roughness when a route needs a value.

4.4 Scoring a route

1. The route from the router is cut every 15 m.
2. For each piece the value of the cell it is in (or a neighbouring cell, for GPS jitter) **for the direction of travel** is looked up. Pieces with no value count as average roughness, and are counted as unknown.
3. A penalty in seconds is added per piece: its time × your Smooth-slider weight × 3 × how far its score is above 0.35.
4. Known obstacles within 12 m add time too: 8 s for each speed bump, about 25 s for each hole or cover (scaled by how often it is hit, and halved on a very rough road), multiplied by the slider weight.
5. Cost = travel time + penalties. Routes that are almost the same (more than 93% within 25 m of each other) are merged, the rest are ranked by cost.

The cards show time, distance, the share of the route in each level, bumps and holes, and the kilometres that are rough and very rough.

4.5 Finding bumps, holes and manhole covers

Single knocks are not part of the road score; they are stored as **events**. An event is a knock that is higher than a bar that grows with speed (about 1.5 m/s^2 at 20 km/h and 3.4 m/s^2 at 70 km/h) and with the square root of the car's stiffness, so a stiff car, which feels everything, needs a bigger knock. For each event the app:

- pairs the knocks of the **front and rear axle** (one wheelbase apart at the speed driven), which tells a real obstacle from a single jolt and gives its length;
- works out the **height** from the movement (a speed bump is 0.1 to 20 m long and 2 to 30 cm high);
- uses the **gyroscope** to see whether the car rolled (one side hit: a hole or cover on the left or right) or pitched (both wheels: a bump), and which side;

- drops knocks in the first and last 90 seconds of a trip at low speed, or in sharp turns (car parks, ramps, kerbs), and anything while the phone was moving or vibrating;
- notices when you **steer round** something: a quick move of 0.3 to 2.2 m sideways and back, on a straight road, from the gyroscope.

Events from many drives are then grouped by place. An obstacle that is **hit on every pass** cannot be avoided: it is a speed bump (or a long table, counted as two). One that is hit **only on some passes**, or that drivers steer round, can be avoided: a hole or cover. One seen once stays *unconfirmed* until it is confirmed by another drive. Only the biggest 30% of holes are shown. After a very deep hit the app asks you to confirm for three seconds (no answer means no).

You can correct any obstacle (*speed bump*, *hole / manhole*, *nothing there*) and mark new ones with the gold buttons, at the position of the car at that moment. **Your labels always win** over what was detected, over OpenStreetMap's known speed bumps and over what the server knows; you can switch each of those sources off.

4.6 How a Loop is planned

1. **Size.** From the time or distance asked, the planner works out the size of six triangle-shaped round trips (start → A → B → start), one for each compass sector, and later a second ring of six rotated by 30 degrees when it needs more choice. For a time target it first assumes about 47 km/h and learns the real ratio from the router's answers; loops that miss the target are resized and asked again, within a budget of router requests.
2. **Over known roads.** The recorded cells are a graph: from a cell you may go to a neighbouring cell (or two steps along an axis, because 1 Hz GPS can skip a cell) that was recorded. The planner finds the corners among the cells *reachable* from the start, and runs a shortest-path search (A*) over the graph for each of the three legs, each leg paying six times more for cells an earlier leg used, and each light, crossing, toll, bump or hole you want to avoid adding 250 m of “cost” to the cells around it. The path is reduced to its corners plus points so that no gap is too long, and these become the via points of the router, which draws the real road geometry.
3. **There and back.** It also tries loops that go out along recorded roads and come back along the same via points, in the directions where recorded roads reach farthest. Both directions of a road count as known, so these are the loops with the least unknown road possible.
4. **Exactly where you are.** The router snaps the first and last points to a road. Both ends are put back on your position, with the short stub added to distance and time. A loop that starts or ends more than 1 km from you is discarded.
5. **Counting.** Every loop is scored like a route. Lights, zebra crossings, tolls and level crossings come from OpenStreetMap (within 18, 14, 25 and 18 m of the route, counted once per junction), bumps and holes from the obstacle map.
6. **Ranking, in strict order of priority:** (1) known road: loops with at most 5% unknown road first, the more known the better; (2) the smoothness you asked for; (3) the time or distance within your tolerance; (4) the fewest of the things you chose to avoid. Loops outside the tolerance only fill up to three results; if there are at least three loops of 95% known road, no other loop is shown; and the loop with the least unknown road is always among the results.
7. **Explorer on the Route tab** (v0.9.15; the share is set in Options → Route Explorer since v0.9.16): you choose from 10% to 100% of the ride that must be on roads **never recorded, with any car, in any direction** (a road counts as recorded if a cell next to it has data). The phone takes the router's alternatives and detours, measures the share of new road of each (every 15 m), and keeps those with at least the chosen share, fastest first. If none has enough it asks the router for up to 10 more routes through places where nothing is recorded; if the road network still cannot give that much, it says so and shows the closest. This is always worked out on the phone, because “never recorded” refers to your own recordings.

8. **Explorer** (loops) does the opposite on purpose: no search over known roads, corners not pulled onto recorded roads, and the ranking prefers unrecorded road.

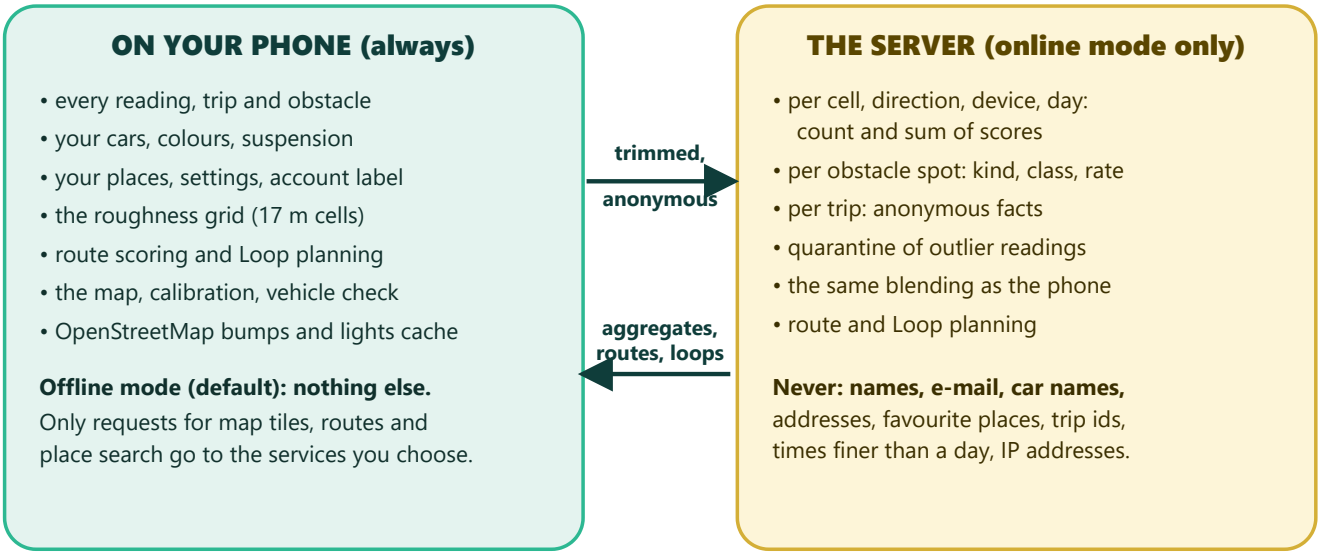
4.7 Checking your vehicles against each other

With more than one vehicle, the app looks at every stretch driven by two or more of them and compares their calibrated values on a logarithmic scale. A vehicle is suspect when, over at least 25 shared stretches, it is off by at least 42% in the same direction on at least 70% of them. With two vehicles, the one that was not calibrated with the guided calibration, and then the one with fewer readings, is named as the most probable. The app then offers to calibrate it, because its readings add bad data.

4.8 Everything that is stored on the phone

Readings (position, time, speed, accuracy, scores, heading, texture and the car), trips, obstacle events and your labels, the OpenStreetMap bump and signal cache, cars and calibrations, places and settings, and the account label. It can all be exported to one file, and deleted.

5. What is shared with others, and how



By default nothing is shared: the app is in **offline mode**. The only requests that leave the phone are for map tiles, route geometry and place search to the services you choose in Options (OpenStreetMap, MapTiler, Stadia, OSRM, openrouteservice, GraphHopper, Nominatim), and for known bumps and traffic lights (Overpass), which see the area needed for the request, not who you are.

If you switch **Online mode** on (after a dialog that explains all of this) the app sends, after each finished trip:

- **The readings of the trip**, with the **first and last 300 m cut off** (so home and work are not in the data), positions rounded to 5 decimals (about 1 m), heading, the score **already converted with your car's calibration** (so cars agree, and a later recalibration does not change what was sent) and only the **day**, never the time. Trips shorter than about 800 m send nothing, and an obstacle within 300 m of any trip start or end stays on the phone.
- **Your own marks**: the speed bumps and holes you marked with the two gold buttons (position, kind and day). Nothing the app detects on its own, and nothing from OpenStreetMap, is sent.
- **Anonymous facts about the trip:** the day, app and Android version, country and language, the suspension setting, the number of vehicles in use (cars with more than 5 km in the last 3 months, never their names), the options that are set (from a fixed list, never addresses, places, e-mail, keys or the server address), how often each feature was used since the last trip (counters only), and a few numbers about the trip (distance, minutes, number of readings and events, average speed and the share of each smoothness level).
- **A random identifier** made on the phone: not your advertising id, not derived from your e-mail address. The server stores only a keyed hash of it.
- **A nickname**, only if you choose one (v0.9.19), for the explorers ranking: it is shown to everyone who opens the ranking, together with the kilometres of new road you registered first. With each trip the app also tells the server whether the car was calibrated (yes or no), which decides whether your roads can count for the golden steering wheel (v0.9.20).

Never sent: your name, any e-mail address (the app has none), your cars' names, home, favourite places or search texts, your API keys, the first and last 300 m of a trip, the time of day, or anything that links the readings of one trip together on the server (each upload part only carries a per-device batch number, so that a retry is never counted twice).

What comes back is only aggregate: routes and loops planned on everyone's data, and (for a future community layer) aggregated cells and obstacles. Nothing about who drove where or when.

Your control: *Send now* and *Delete my data on the server* in Options; deleting removes everything the device ever sent, including the trip facts. Uploads are retried safely (each part has a batch id), the phone shows what failed, and you can keep earlier recordings on the phone only.

6. What the server does with it

6.1 What it stores

- **Cells:** for each road cell (about 17 m), driving direction (45-degree bin), device (a keyed hash) and day: how many readings and the sum of their (calibrated) scores. No trips as such (only a per-device batch number that makes a retry harmless), no routes, no start or end points, no times finer than a day.
- **Obstacles:** for each obstacle spot, kind and device: position, how the phone classified it, how often it is hit.
- **Trip facts:** one row per trip with the anonymous facts above, kept for the developer's understanding of how the app is used; a script summarises them.
- **Quarantine:** the (cell, direction, device) combinations and the devices whose values are far from the others (section 6.3).
- Nothing else: no e-mail, no name, no phone model, no car name, no address, no IP address (the web server in front does not log them).

Every night the server deletes data older than two years and keeps only the newest 20 passes per cell and direction (the blend uses 10), so the database stays small. A device can ask to erase everything it ever sent.

6.2 How the smoothness of a cell is worked out

The server uses the **same rule as the phone**, with devices instead of cars: for a cell and direction, each device's passes on each day are one “drive”; the latest drive counts fully, earlier ones fade with $e^{-\text{days older} / 30}$, and only the last 10 drives count. Because the phones already send calibrated scores, values from different cars and phones are comparable. A cell exists for the outside world only when at least **min_contributors** different devices recorded it (the setting is 1 while testing, and can be raised so that a road is only used when several people agree).

6.3 Quarantine: keeping bad data out

A reading that is far from the middle of the normal curve for its stretch of road almost always comes from a car that is not calibrated, or a phone that is not firmly mounted. For every cell and direction recorded by at least **three** devices, the server compares each device's average with the **median** of all of them, on a logarithmic scale (a wrong calibration multiplies the score). A device is an **outlier** for that cell when it is farther from the median than the larger of a factor of 1.65 and three robust standard deviations (the median absolute deviation × 1.4826). Outlier values are listed in a quarantine table and are **left out of the averages** that everybody gets. A device that is an outlier on at least half of the 20 or more cells it shares with others, or whose typical deviation is a factor of 1.57 or more, is set aside as a whole until a later check finds it in line again. The cells an upload touches are judged when it arrives, and everything is judged again every night. Nothing is deleted, so data comes back when the picture changes.

6.4 Obstacles

For each spot of about 17 m and each kind (speed bump, hole or cover), the server looks at what the devices reported. A spot exists only when enough different devices reported it. Its position is the average of the reports, its class is **unavoidable** when at least half of the reports say so and **avoidable** otherwise, its “rate” (how often it is hit) is the average, and the number of contributors is kept. Reports from quarantined devices are not used. A phone can ask for routes without other people's bumps, or without their holes, and the server then leaves that kind out.

6.5 Planning routes and loops for a phone

The server runs the **same planner as the phone** (a Python port of the route scoring and the Loop planner, section 4.4 and 4.6) on the combined data, with an OSRM router for the road geometry and OpenStreetMap data for traffic lights, zebra crossings, tolls and level crossings (imported from an OpenStreetMap extract). A route request carries the start, the destination, the smoothness weight and the choice of obstacle sources; a loop request carries the start, the time or distance, the tolerance, the weight, the things to avoid and the Explorer switch. It loads the cells around the route or loop (only those with enough contributors and not in quarantine), plans, scores and ranks, and returns the best.

6.6 What it replies

- **Routes and loops:** for each, the geometry as a series of points every 15 m with the smoothness of each piece (or “unknown”), the time, distance and cost, the share of known road, the average roughness, the rough and very rough kilometres, the number of bumps and holes, and for a loop also how well it fits the time or distance, whether it is within the tolerance, the share of road driven twice, the count of each thing to avoid, and whether the 5% unknown rule could be met. The phone shows them exactly as it shows its own, and adds your own bump and hole marks when you switched the server's off.
- **Aggregated cells:** latitude, longitude, direction, value and number of contributors per cell and direction; and **aggregated obstacles**: position, kind, class and contributors. No times, devices or trips.
- **Status**, with the number of loops it returns and the size of the quarantine.

If the server cannot be reached, the phone plans on its own data and says so.

6.7 How it protects people

Devices are known only by a keyed hash of a random id; requests carry a token that only that id can produce; there are limits on upload size and on readings per device per day; and the first and last 300 m of every trip never leave the phone, so the data cannot show where anyone lives or works. The server keeps no access log with addresses.

7. Numbers at a glance

What	Value
Minimum speed for smoothness readings	5 km/h
GPS accuracy accepted	20 m or better
Road cell	about 17 m, per direction (45 degrees)
Drives blended per cell	last 10; latest 100%, earlier $e^{-\text{days}/30}$
Levels	0 to 5: Smooth 4–5, Good 3–4, Fair 2–3, Rough 1–2, Very rough 0–1
Suspension scale shown	1.0× to 6.5×, normal 2.5×
Route scoring step	15 m
Speed bump / hole penalty in routes	8 s / about 25 s × weight
Loop length	10 min to 2 h, or up to 200 km (125 mi), tolerance 5–25%
Loop known-road rule	at least 95% known road whenever possible
One thing to avoid is worth	250 m of detour in the search
Loop ends from your position	put back exactly; discarded beyond 1 km
Auto-stop	30 minutes without moving
Phone-moved pause	net turn above about 50° in 1.5 s, quiet for 4 s after start or resume
Vehicle check	25+ shared stretches, 42% off, 70% the same way
Upload trimming	first and last 300 m; trips under about 800 m not sent
Server quarantine	3+ devices; factor 1.65 or 3 robust sigmas
Server retention	two years; newest 20 passes per cell and direction

8. Honest limits

- The score is an **estimate** from a phone's sensors. It is only as good as the mount, the calibration and the number of drives behind a road, and it says nothing about safety.
- Roads nobody has driven are unknown: a Loop can only be built over what is known, which is why Explorer exists.
- Vibration from other apps cannot be stopped by an app; the motor-vibration detector needs a sensor rate of about 300 Hz or more, and a phone with slower sensors cannot tell it from the road.
- A real router sometimes adds a short detour near a via point, so a loop's unknown share can be a little higher than planned (the cards show it).
- Please drive safely: set the app up before you set off and never handle the phone while driving.